Project assignment:	Reorganisation of an outpatient department in a hospital
Course:	Simulation modelling and analysis (F000941)
Year:	2025-2026

In this assignment, we consider the construction of a cyclic appointment schedule for an outpatient radiology department with a single server. The goal of this study is to minimize the waiting times for urgent and elective patients while considering the impact on the daily overtime. The cyclic appointment schedule consists of time slots dedicated to urgent or elective patients only. Elective patients are patients that have made an appointment and are planned beforehand. Urgent patients are patients that have entered the hospital without an appointment and should be treated on the same day. The construction of a cyclic appointment schedule involves a strategic decision related to the capacity distribution and a tactical decision related to the timing of the time slots. In order to evaluate the performance of the cyclic appointment schedule, the evaluation is performed on the operational level and operational variability should be taken into account.

1. Strategic and tactical decisions

The current cyclic appointment schedule in the outpatient department is shown in Figure 1. These slots form a time slot schedule for one week, which is repeated over time in a cyclic manner. Patients can be scheduled from Monday to Saturday, not on Sunday. Note that on Thursday and Saturday the department is only open for half a day. Each day, also called a session, is divided into time slots of equal length during which one patient is served at a time. Each time slot has a duration equal to the expected appointment duration, i.e. 15 minutes. The capacity of a time slot is reserved only for patients of the related patient type. We assume that all resources, such as nurses and doctors, are available. The hospital currently uses a cyclic slot schedule with 146 elective slots and 14 slots for urgent patients indicated by the slots in grey in Figure 1.

The goal of this study is to redesign the cyclic appointment schedule by investigating two decisions, i.e.

- The capacity distribution between time slots dedicated to urgent patients and elective patients: In the current cyclic schedule (cf. Figure 1) there are 14 time slots dedicated to urgent patients. The hospital wants to revise the distribution of the time slots. The department head indicated that the number of time slots dedicated to urgent patients should range between 10 and 20 time slots. These decisions impact the waiting times of both elective and urgent patients. Installing too many slots for urgent patients leads to a smaller number of time slots open for elective cases and, hence, to longer appointment waiting times for these latter patients.
- The timing of slots for urgent patients over the planning horizon: In this respect, the hospital wants to compare the following strategies:
    - Strategy 1: Currently, the hospital has installed the time slots for urgent patients at the end of each morning or afternoon block as indicated in Figure 1. Increasing the number of slots for urgent patients will increase or decrease this number at the end of each morning or afternoon block. Time slots for urgent patients are distributed evenly over the morning and afternoon block and over the days.

- Strategy 2: The time slots dedicated to urgent patients are evenly distributed over the day taking the opening hours of the department into account. When 16 or 32 slots are available on a particular day, the distribution of slots over the day is illustrated in Appendix (Figure 5).
- Strategy 3: Starting from the beginning of a session, a time slot for urgent patients is planned each time after a block of six time slots for elective patients. This strategy is illustrated in Appendix (Figure 6).

Note that these strategies try to divide the number of slots evenly over the days as much as possible taking the opening hours into account.

Slot nr	Start time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Slot 1	08:00							
Slot 2	08:15							
Slot 3	08:30							
Slot 4	08:45							
Slot 5	09:00							
Slot 6	09:15							
Slot 7	09:30							
Slot 8	09:45							
Slot 9	10:00							
Slot 10	10:15							
Slot 11	10:30							
Slot 12	10:45							
Slot 13	11:00							
Slot 14	11:15							
Slot 15	11:30							
Slot 16	11:45							
		Lunch break						
Slot 17	13:00							
Slot 18	13:15							
Slot 19	13:30							
Slot 20	13:45							
Slot 21	14:00							
Slot 22	14:15							
Slot 23	14:30							
Slot 24	14:45							
Slot 25	15:00							
Slot 26	15:15							
Slot 27	15:30							
Slot 28	15:45							
Slot 29	16:00							
Slot 30	16:15							
Slot 31	16:30							
Slot 32	16:45							

Figure 1. Current cyclic appointment schedule in the outpatient department

2. Operational decisions

In order to determine the performance of a cyclic appointment schedule accurately, operational decisions related to the scheduling of individual patients should be included in the analysis. As a result, individual patient arrivals and scan durations, are simulated to evaluate the impact of the cyclic appointment schedule on the waiting times and overtime.

2.1 Operational execution of the cyclic schedule and performance measures

In the operational phase, elective patients call the front desk of the outpatient department during the opening hours in order to make an appointment. These patients are given an appointment in the order they arrive (First Come First Serve), i.e. the next elective patient that calls the outpatient department is appointed to the next available time slot dedicated to elective patients in the planning horizon. In addition, the patient is given an appointment time, i.e. the time the patient should show up at the department, which can be different from the start time of the time slot appointed to the patient. The hospital has currently a simple FCFS appointment system implemented, which implies that the appointment time is equal to the start time of the time slot appointed to the patient. However, the hospital wants to experiment with other appointment scheduling rules (cf. infra). The appointment waiting time of an elective patient is the time between their request for an appointment (phone call) and their actual appointment time. The scan waiting time of an elective patient time is the time between the physical arrival of the patient in the department and the start time of the treatment. Both these concepts are illustrated for an elective patient in Figure 2.

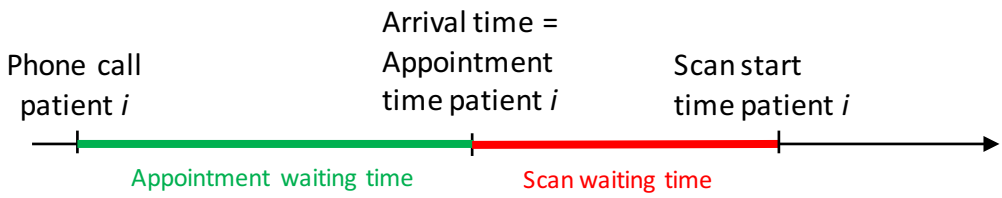

Figure 2. Illustration of the appointment waiting time and the scan waiting time for an elective patient.

Urgent patients are non-elective patients that arrive to the outpatient department without having an appointment. Given the urgency for these patients, the patients should be examined the same day. These urgent patients are assigned to the next available time slot dedicated to urgency patients. If there is no time slot left on a particular day, these patients are assigned to overtime time slots, i.e. time slots after the normal opening hours of the outpatient department when all the elective patients have been served. The time working during the lunch break is not accounted as overtime. No urgent patients are treated during the lunch break. As a result, the urgency patients do not have an appointment waiting time but only a scan waiting, defined as the time between the physical arrival of the patient in the department and the start time of the treatment. This concept is illustrated for an urgency patient in Figure 3. Note that urgent patients arriving outside the opening hours of the outpatient department (e.g. on Thursday or Saturday afternoon, Sunday or at night) should not be considered as they are examined in the emergency department of the hospital.

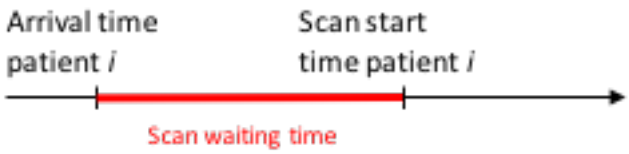

Figure 3. Illustration of the scan waiting time for an urgency patient.

2.2 Performance measures

In order to evaluate a cyclic appointment schedule, the following performance measures are utilized:

- The appointment waiting time of elective patients
- The scan waiting time of elective patients
- The scan waiting time of urgency patients
- The overtime required in the department to examine all patients. The daily overtime can be calculated by calculating the difference between the scan end time of the last patient on a particular day and the session end time of the day.

However, the hospital manager envisages a hierarchy between the different objectives and primarily wants to focus upon the appointment waiting time of elective patients and the scan waiting time of urgency patients as these seem the most important objectives when determining the capacity distribution and the timing of the time slots for urgent patients. In order to find a suitable objective function, the waiting time is normalized by the maximum desired waiting time postulated by the management for the different patient types, i.e. 9 hours for urgent patients and 1 week or 168 hours for elective patients, leading to the respective weights $w_u = 1/9$ and $w_e = 1/168$. Hence, the objective is to find the design that minimizes the weighted waiting time, i.e.

Minimize $w_e Wt_e + w_u Wt_u$

With w_i priority weight of patient type i
Wt_i average waiting type for patient type i

The other two objectives, i.e. the scan waiting time of elective patients and the overtime, are secondary objectives. The findings related to these objectives are only used to break ties between different appointment systems when necessary.

2.3 Appointment scheduling rules

As indicated, the hospital applies currently an FCFS appointment system where the appointment time is equal to the assumed time of service, i.e. the start time of the time slot assigned to the patient. However, in the literature different other single-pass appointment scheduling rules exist. Some appointment scheduling rules give patients an earlier or later appointment time instead of the actual start time of their slot. In the simulation experiment, the hospital would like to experiment with different appointment scheduling rules. Each rule schedule the patients one by one in the order of first come first serve (FCFS), resulting in a time slot allocation. However, the different rules assign a different appointment time to the patients. In this study, we examine the impact of variations of the following appointment rules, i.e.

- **Rule 1 – Plain FCFS rule:** The appointment time is equal to the start time of the assigned time slot.
- **Rule 2 - Bailey-Welch rule:** The appointment time of the first *K* patients of a session is equal to the start time of the first time slot of the particular day. All subsequent patients receive an appointment time equal to the slot start time of the patient minus

the expected scan duration, which is equal to the length of one time slot. You can assume *K* is equal to 2.

- **Rule 3 - Blocking rule:** A session is split up in blocks of multiple time slots. All patients of whom the time slot lies within the same block get an appointment time equal to the start time of the block. This rule can be altered by assuming a different number of slots per block (*B*). You can assume *B* is equal to 2.
- **Rule 4 - Benchmarking rule:** All patients are assigned to either an earlier appointment time than the start time of their slot, i.e. appointment time = start time of appointed time slot minus $k_\sigma \times \sigma_e$ (with σ_e the standard deviation of the scan duration of elective patients and k_σ designates how much the possible deviations in the scan duration are taken into account. You can assume k_σ is equal to 0.5.

2.4 Operational variability

The performance of the outpatient department is subject to operational variability. We assume that there are two types of variability related to the patient arrivals and the service times. Patients arrive unexpectedly and need an appointment within a short time span. On the other hand, there is also some processing variability as the process includes a lot of human activity. The appointment duration is therefore inherently variable.

In the following, we describe the characteristics of the operational variability for each type of uncertainty by means of a probability distribution with adequate parameters.

2.4.1 Uncertainty of patient arrivals

The operational variability impact the arrival of elective and urgent patients and accordingly the associated waiting times for both types of patients.

Elective patients - The variability in the arrival process for elective patients is related to both the phone call time of the patients and their actual arrival time at the moment of appointment. The distribution of the phone call times of elective patients is estimated via the phone logs of the front desk of the hospital. This desk is only open from Monday to Friday between 8 am and 5 pm. Hence, no elective patients can call on Saturday or Sunday. The call time of patients can typically be generated from a memoryless negative exponential distribution to generate the interarrival times between patient arrivals. When fitting the empirical data, we find that the number of elective arrivals per day follows a Poisson distribution with λ_e = 28.345. No difference has been found in the arrival pattern between different hours of a day. The variability of the actual arrival time at the moment of appointment refers to no-shows and the unpunctuality of patients, which is the difference between the patient appointment time and the actual arrival time (cf. Figure 4). Patients can arrive early or late in comparison to their appointment time. Patients arriving early may lead to the congestion of the waiting area. In case a patient arrives late, the appointment schedule for the rest of the session may be impacted. In this study, we assume that the tardiness distribution function follows a normal distribution with a mean of 0 (i.e. the patients are expected to arrive on time) and a standard deviation of 2.5 minutes and that the probability of not showing up is equal to 2%.

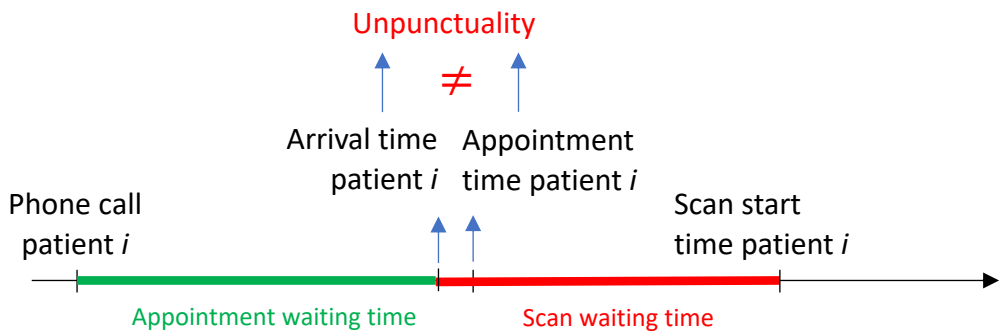

Figure 4. Illustration of the unpunctuality (arrival time <> appointment time)

Urgent patients - The variability in the arrival of urgent patients is related to the arrival time of these patients, which is equal to their moment of actual appearance at the outpatient department. The average amount of urgent patients per day has been estimated by the department personnel due to the lack of historical data. The arrival rate of urgent patients is generated by the negative exponential distribution with λ_u = 2.5 patients per entire day the outpatient department is open and λ_u = 1.25 patients per day the outpatient department is only open for half a day. As there are significant differences between the distributions of the scan duration for different examination types due to setup times, the distribution of the required scan types for urgent patients has been analyzed and modelled by a discrete distribution. The frequency distribution of the most common urgent patient categories has been indicated in Table 2.

2.4.2 Uncertainty of service times

The operational variability may impact the actual duration of the service encounter for a patient. As a result, the appointment and scan waiting times of subsequent elective and urgent patients is impacted. To estimate the scan duration distributions, we analysed data logs of the department and conducted in-dept interviews with personnel arising to the following considerations, i.e.

- The department data logs showed that for elective patients there are no significant differences between the distributions of different scan types. Consequently, only one scan duration distribution was estimated for all elective patients. The elective scan duration follows a normal distribution with a mean of 15 minutes and a standard deviation of 3 minutes.
- The department data logs showed that for urgent patients there are significant differences between the distributions of the scan durations between different scan types due to setup times. The durations of all scan types for urgent patients are estimated to follow a normal distribution. Table 2 contains the mean and the standard deviation of the scan duration distribution for the most common scan types for urgent patients.

Scan Type	Freq. (%)	Scan duration	
		Mean (min)	St dev (min)
Brain	70%	15	2.5
Spine			
Lumbar vert	10%	17.5	1
Cervical ver	10%	22.5	2.5
Abdomen MI	5%	30	1
Others	5%	30	4.5

Table 2. Frequency distribution and scan duration distribution for urgent patients.

3. Assignment

The department would like to benchmark the appointment system that is currently used to schedule individual patients with alternative configurations leading to the best interrelated decision regarding to the capacity allocation between patient types, the allocation of time slots throughout the cyclic horizon and the appointment scheduling rule.

Appendix. Illustration of the timing strategies for scheduling time slots for urgent patients

		# Time slots for urgent patients			
Slot nr	**Start time**	**1**	**2**	**3**	**4**
Slot 1	08:00				
Slot 2	08:15				
Slot 3	08:30				
Slot 4	08:45				
Slot 5	09:00				
Slot 6	09:15				
Slot 7	09:30				
Slot 8	09:45				
Slot 9	10:00				
Slot 10	10:15				
Slot 11	10:30				
Slot 12	10:45				
Slot 13	11:00				
Slot 14	11:15				
Slot 15	11:30				
Slot 16	11:45				
		Lunch break			
Slot 17	13:00				
Slot 18	13:15				
Slot 19	13:30				
Slot 20	13:45				
Slot 21	14:00				
Slot 22	14:15				
Slot 23	14:30				
Slot 24	14:45				
Slot 25	15:00				
Slot 26	15:15				
Slot 27	15:30				
Slot 28	15:45				
Slot 29	16:00				
Slot 30	16:15				
Slot 31	16:30				
Slot 32	16:45				

Figure 5a. Time slot distribution according to Strategy 2 (32 slots per day)

		# Time slots for urgent patients			
Slot nr	**Start time**	**1**	**2**	**3**	**4**
Slot 1	08:00				
Slot 2	08:15				
Slot 3	08:30				
Slot 4	08:45				
Slot 5	09:00				
Slot 6	09:15				
Slot 7	09:30				
Slot 8	09:45				
Slot 9	10:00				
Slot 10	10:15				
Slot 11	10:30				
Slot 12	10:45				
Slot 13	11:00				
Slot 14	11:15				
Slot 15	11:30				
Slot 16	11:45				

Figure 5b. Time slot distribution according to Strategy 2 (16 slots per day)

		# Time slots for urgent patients			
Slot nr	**Start time**	**1**	**2**	**3**	**4**
Slot 1	08:00				
Slot 2	08:15				
Slot 3	08:30				
Slot 4	08:45				
Slot 5	09:00				
Slot 6	09:15				
Slot 7	09:30				
Slot 8	09:45				
Slot 9	10:00				
Slot 10	10:15				
Slot 11	10:30				
Slot 12	10:45				
Slot 13	11:00				
Slot 14	11:15				
Slot 15	11:30				
Slot 16	11:45				
		Lunch break			
Slot 17	13:00				
Slot 18	13:15				
Slot 19	13:30				
Slot 20	13:45				
Slot 21	14:00				
Slot 22	14:15				
Slot 23	14:30				
Slot 24	14:45				
Slot 25	15:00				
Slot 26	15:15				
Slot 27	15:30				
Slot 28	15:45				
Slot 29	16:00				
Slot 30	16:15				
Slot 31	16:30				
Slot 32	16:45				

Figure 6. Time slot distribution according to Strategy 3